

Self-Attention (Q, K, V)

DATA 443/643 · Week 3, note 3.2

After CMU 11-711: scaled dots, mix values, then a causal mask.

Lecture plan

- 1 Query asks, key is asked, value is mixed
- 2 Scale by sqrt of d_k , then softmax
- 3 Multi-head: concat, then W_O
- 4 Causal mask: minus infinity, not zero

First time: what / why / formula

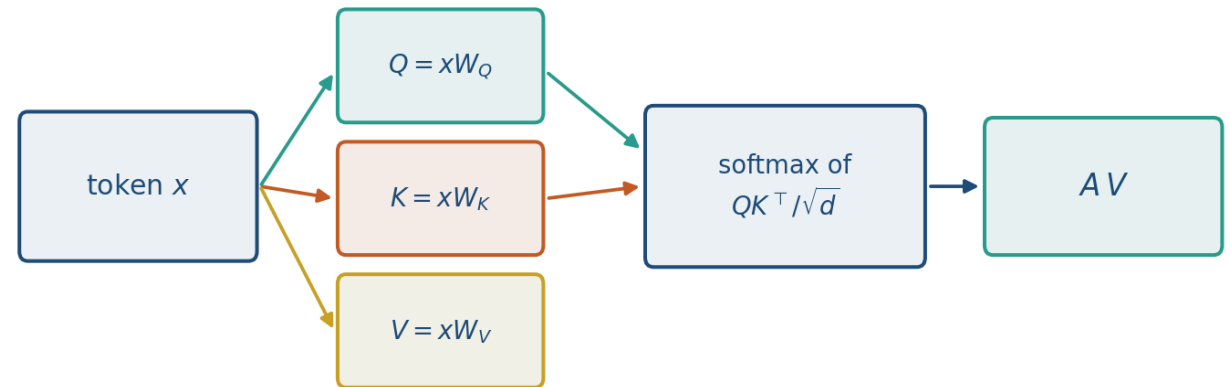
- What / why. A weighted sum of values, with weights from queries compared to keys. Every token plays all three roles. RNNs mix left-to-right through h_t . Self-attention mixes any pair in one layer, in parallel.
- Architecture. $X \mapsto Q, K, V$ by three matrices. Scores $QK^\top / \sqrt{d_k}$, softmax over keys, mix V . Multi-head: several of those in parallel, then W_o . Causal mask for LMs.
- Formula. $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})V$.
- Tradeoffs. + Parallel, path length 1, multi-head specialization. – $O(T^2)$; mixing keys instead of values is a common bug; no positions until note 3.3.

How. Lab 3: one head by hand, then a causal mask. Zero is the wrong mask fill; use $-\infty$.

Query, key, value

- Q, K, V are three linear maps of X.
- A row of QK transpose: how much this token wants each position.
- Then mix values. Mixing keys is the lab bug.

Query asks, key is asked, value is what you mix in



Scaled dot-product attention

$$A = \text{softmax}\left(QK^T / \sqrt{d_k}\right), \quad \text{out} = AV$$

Variance of a dot product grows with d_k .

The scale keeps softmax from becoming one-hot.

Softmax over keys (last dim), not over queries.

A 2 by 2 you will write

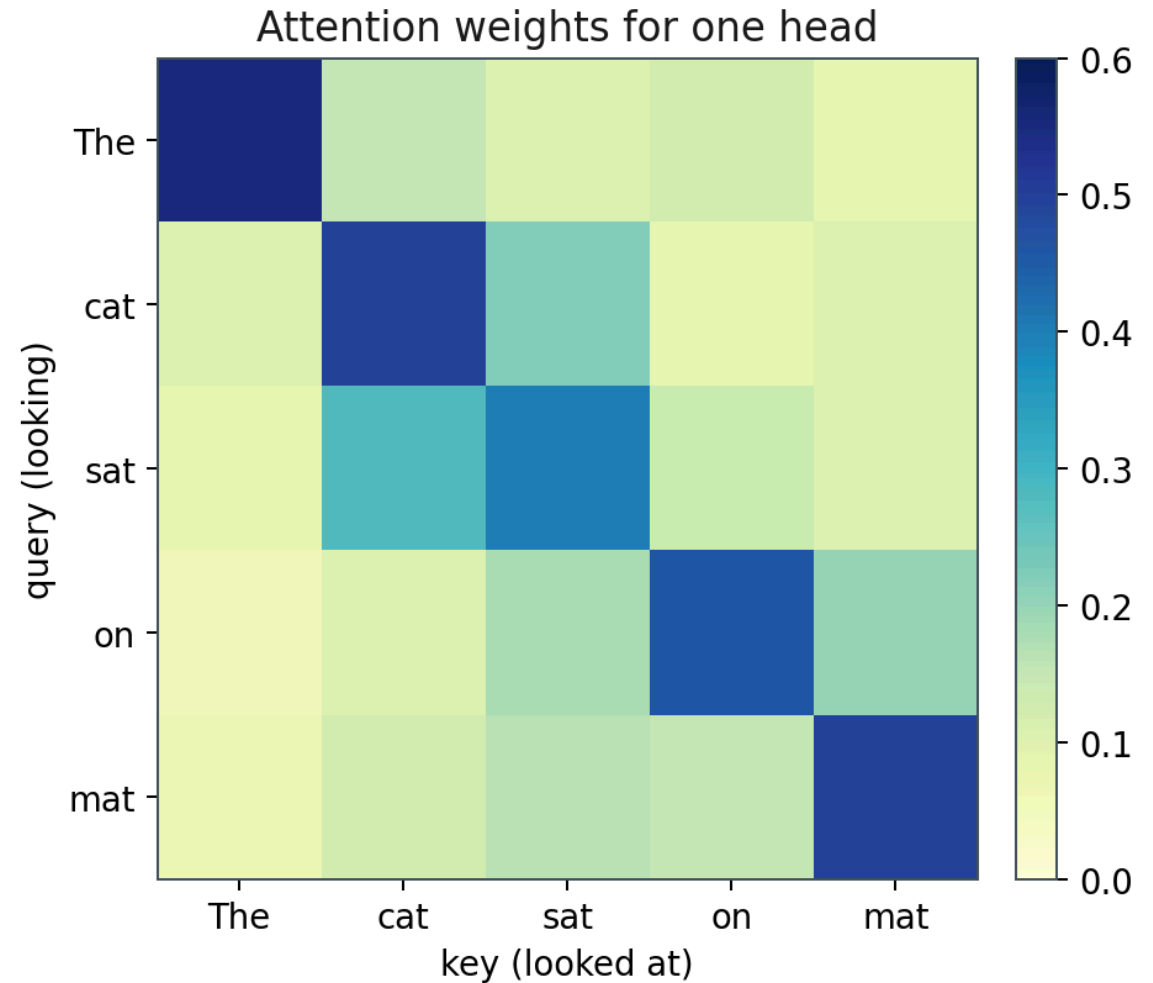
$S = \text{diag}(2, 2)$, $d_k = 2$. Scale, softmax, mix values.

- $S = \text{diag}(2, 2)$, $d_k = 2$.
- Row 1 softmax is about [0.80, 0.20].
- If $V = I$, the output is A .

row	scaled scores	softmax A	mix of $V = I$
1	[1.41, 0]	[0.80, 0.20]	[0.80, 0.20]
2	[0, 1.41]	[0.20, 0.80]	[0.20, 0.80]

A row is a distribution

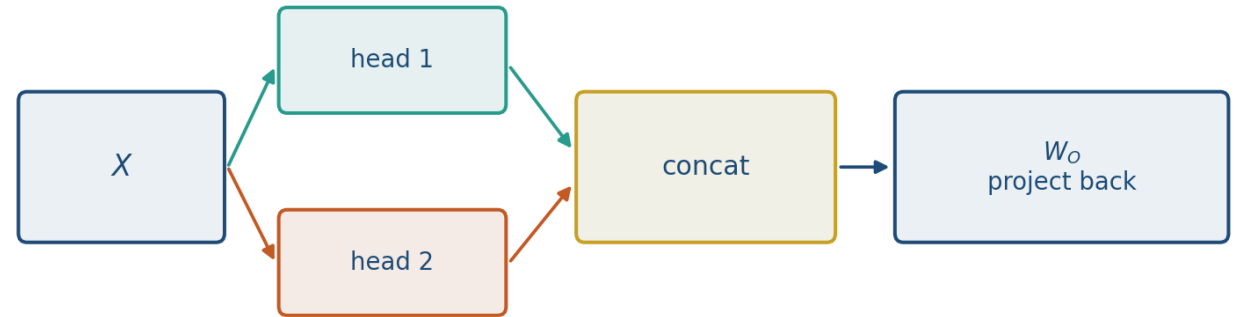
- Large mass on a name: this pronoun just looked there.
- Equal keys split mass equally.
- Lab 3 heatmaps this for $T = 4$.



Multi-head

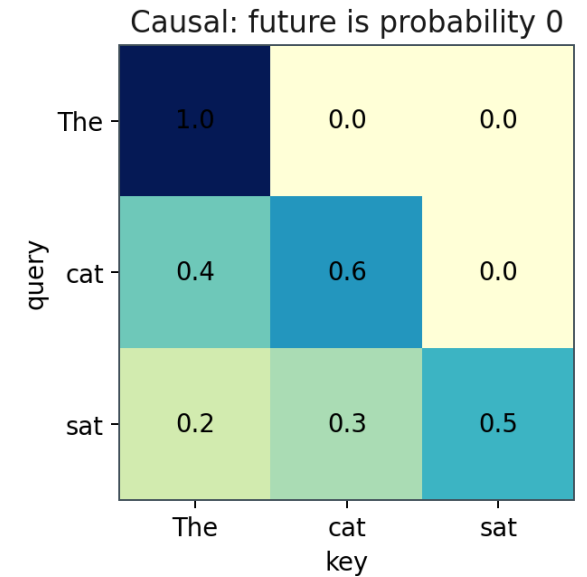
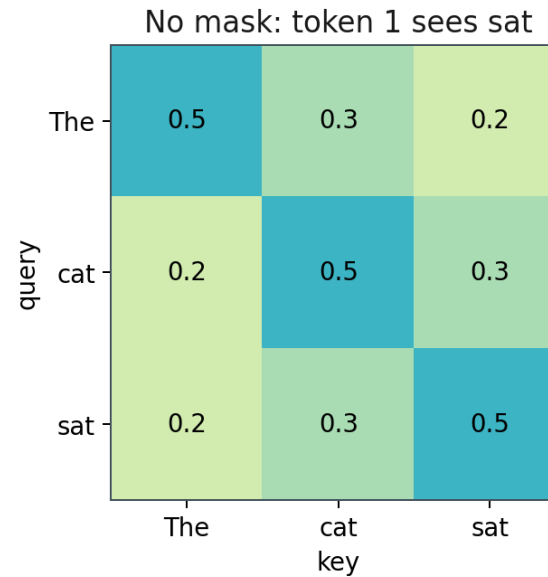
- Several small attentions in parallel.
- Concatenate, then one linear W_O .
- Heads can specialize. You still compute one by hand.

Multi-head: several small attentions, then one linear map



Causal mask

- Illegal future scores become minus infinity.
- Softmax of a zero is not probability zero.
- This mask is GPT. BERT hides tokens instead.



What to take from this lecture

Write it

$A = \text{softmax}(QK^T / \sqrt{d_k})$,
then AV .

Scale it

Skip the square root and the row
becomes too peaked.

Mask it

Lab 3: first bidirectional, then GPT-
style.